MATH 111 Calculus 1, NUU, FALL 2026

Dr. Salnikova Chekeres **Tutorial 2**

Limits: Chapter 1, Sections 1.5–1.7 (the limit of a function; calculating limits; the precise definition).

Key facts to have at hand

- $\lim_{x\to a} f(x) = L$ means: $f(x)$ can be made **as close to** L **as we like** by taking x close enough to a, $x \neq a$. The value $f(a)$ — defined or not — plays no role: a limit is about the *approach*, not the *arrival*.
- **One-sided limits** $\lim_{x\to a^-} f(x)$, $\lim_{x\to a^+} f(x)$ use only $x < a$, resp. $x > a$. $\lim_{x\to a} f(x) = L$ *if and only if* both one-sided limits exist and equal L.
- **Infinite limits:** $\lim_{x\to a} f(x) = \infty$ means $f(x)$ becomes as large as we like for x close enough to a — a sentence about behaviour, not a number. Then $x = a$ is a **vertical asymptote**. Check the sign on each side separately.
- **Limit laws** (all limits at $x \to a$ assumed to exist): $\lim[f \pm g] = \lim f \pm \lim g$; $\lim[cf] = c\lim f$; $\lim[fg] = \lim f \cdot \lim g$; $\lim \frac{f}{g} = \frac{\lim f}{\lim g}$ *provided* $\lim g \neq 0$; $\lim[f]^n = [\lim f]^n$; $\lim \sqrt[n]{f} = \sqrt[n]{\lim f}$ (for even n: $\lim f > 0$).
- **Direct substitution:** if f is a polynomial or a rational function and a is in its domain, $\lim_{x\to a} f(x) = f(a)$.
- $0/0$ **is an instruction, not an answer:** rewrite, then substitute — factor and cancel; multiply by the conjugate; combine fractions; go one-sided when the formula changes at a (absolute values, cases). Two functions that agree for all $x \neq a$ have the same limit at a.
- **Squeeze Theorem:** if $g(x) \leq f(x) \leq h(x)$ near a (except possibly at a) and $\lim g = \lim h = L$, then $\lim f = L$.
- **Precise definition:** $\lim_{x\to a} f(x) = L$ means: for every $\varepsilon > 0$ there is a $\delta > 0$ such that if $0 < |x-a| < \delta$ then $|f(x) - L| < \varepsilon$. (ε is the challenge, δ the response; δ may depend on ε; a smaller δ always works too.) $\lim_{x\to a} f(x) = \infty$: for every M there is $\delta > 0$ such that if $0 < |x - a| < \delta$ then $f(x) > M$.

Part I · Reading and calculating limits (Sections 1.5–1.6) · 50 minutes

Warm-up I · True or false? (justify with one sentence or one counterexample)

1. If $\lim_{x\to a} f(x)$ exists, then $f(a)$ is defined.
2. If $f(a)$ is defined, then $\lim_{x\to a} f(x)$ exists.
3. If $\lim_{x\to a} f(x) = L$ and $f(a)$ is defined, then $f(a) = L$.
4. $\lim\limits_{x\to 2} \dfrac{x^2-4}{x-2} = 4$.
5. If $\lim_{x\to a} f(x) = 0$ and $\lim_{x\to a} g(x) = 0$, then $\lim_{x\to a} \dfrac{f(x)}{g(x)}$ does not exist.
6. If $\lim_{x\to a} f(x) = 2$ and $\lim_{x\to a} g(x) = 0$, then $\lim_{x\to a} \dfrac{f(x)}{g(x)}$ is not a real number.
7. $\lim\limits_{x\to 0} \dfrac{|x|}{x} = 1$.
8. If $\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L$, then $\lim_{x\to a} f(x) = L$.
9. If $\lim_{x\to 0} f(x) = \infty$, then $f(0)$ is not defined.
10. $\lim\limits_{x\to 0} x \sin\dfrac{1}{x} = 0$.

Solution.

1 False. A limit can exist where the function has no value. Take $f(x) = \dfrac{x^2-4}{x-2}$: for $x \neq 2$ it equals

$x+2$, so $\lim_{x\to 2} f(x) = 4$; but $f(2)$ is not defined (substituting gives $0/0$). The limit only looks at x *near* 2, never at 2 itself.

2 False. A function can have a value and no limit. The Heaviside function $H(x) = 0$ for $x < 0$, $H(x) = 1$ for $x \geq 0$ has $H(0) = 1$; but from the left the values are 0 and from the right they are 1, so the one-sided limits differ and $\lim_{x\to 0} H(x)$ does not exist.

3 False. Define $f(x) = \dfrac{x^2-4}{x-2}$ for $x \neq 2$ and $f(2) = 7$. Then $\lim_{x\to 2} f(x) = 4$ (the approach) while $f(2) = 7$ (the arrival). Nothing in the definition of a limit connects the two numbers.

4 True. Dividing the numerator and the denominator by $x-2$ is allowed only where $x-2 \neq 0$, that is for $x \neq 2$ — and the limit as $x \to 2$ looks only at such x. So for $x \neq 2$, $\dfrac{x^2-4}{x-2} = \dfrac{(x-2)(x+2)}{x-2} = x+2$, and $x+2 \to 4$ as $x \to 2$.

5 False. "0/0" decides nothing: with $f(x) = x^2$, $g(x) = x$ the quotient is $x \to 0$; with $f(x) = g(x) = x$ it is $1 \to 1$; with $f(x) = x$, $g(x) = x^2$ it is $1/x$, which has no limit. Same "0/0", three different outcomes — that is why $0/0$ is an instruction to do more work, not an answer.

6 True. Suppose the quotient had a real limit Q. Then, by the product law, $\lim f(x) = \lim\left[g(x) \cdot \frac{f(x)}{g(x)}\right] = 0 \cdot Q = 0$ — but $\lim f(x) = 2$. Contradiction. So f/g has no real limit (it is unbounded near a: it may tend to $+\infty$, to $-\infty$, or to neither, depending on the sign of g on each side).

7 False. For $x > 0$, $|x|/x = x/x = 1$; for $x < 0$, $|x|/x = -x/x = -1$. The right-hand limit is 1 and the left-hand limit is -1; since they differ, the two-sided limit does not exist — so it is certainly not 1.

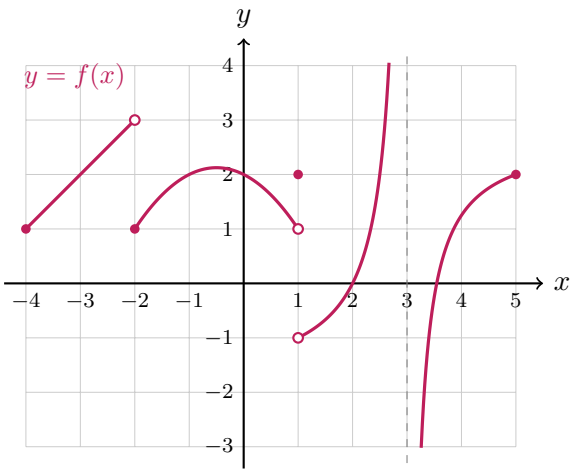
8 True. This is exactly the theorem on one-sided limits: the two-sided limit exists and equals L if and only if both one-sided limits exist and equal L.

9 False. Let $f(x) = 1/x^2$ for $x \neq 0$ and $f(0) = 1$. Then $f(0)$ is defined, and still $\lim_{x\to 0} f(x) = \infty$: an infinite limit describes the values for x near 0, and the value at 0 plays no role.

10 True. Since $|\sin(1/x)| \leq 1$, we have $-|x| \leq x \sin\frac{1}{x} \leq |x|$ for $x \neq 0$. Both $-|x|$ and $|x|$ tend to 0, so by the Squeeze Theorem $x \sin\frac{1}{x} \to 0$.

Problem 1 *Reading limits off a graph.*

The graph of a function f is shown; the dashed line is a vertical asymptote.



Find each of the following. Name the value of the function. Explain.

(a) $\lim_{x\to -2^-} f(x)$ (b) $\lim_{x\to -2^+} f(x)$ (c) $\lim_{x\to -2} f(x)$

(d) $\lim_{x\to 0} f(x)$

(e) $\lim_{x\to 1^-} f(x)$ (f) $\lim_{x\to 1^+} f(x)$ (g) $\lim_{x\to 1} f(x)$

(h) $\lim_{x\to 3^-} f(x)$ (i) $\lim_{x\to 3^+} f(x)$ (j) $\lim_{x\to 3} f(x)$

Solution. How to read the picture: for $\lim_{x\to a^-}$ look at the points of the graph with x slightly *less* than a and see what height they approach; for $\lim_{x\to a^+}$ use x slightly *greater* than a; the two-sided limit exists only if these two heights agree. The *value* $f(a)$ is the filled dot on the vertical line $x = a$ (an open dot is *not* a point of the graph); if there is no dot, $f(a)$ is undefined.

At -2 (a) Coming from the left, the graph is the segment rising towards the open dot at $(-2, 3)$; the heights approach 3: $\lim_{x\to -2^-} f(x) = 3$. (b) Coming from the right, the parabola starts at height 1: $\lim_{x\to -2^+} f(x) = 1$. (c) $3 \neq 1$: the two sides disagree, so $\lim_{x\to -2} f(x)$ does not exist. Value: $f(-2) = 1$ — the filled dot at $(-2, 1)$ belongs to the graph. A value without a limit.

At 0 (d) The curve passes through $(0, 2)$ without a break; from both sides the heights approach 2: $\lim_{x\to 0} f(x) = 2$. Value: $f(0) = 2$. Here value and limit agree.

At 1 (e) From the left the parabola comes down to the open dot at $(1, 1)$: $\lim_{x\to 1^-} f(x) = 1$. (f) From the right the next piece starts at the open dot at $(1, -1)$: $\lim_{x\to 1^+} f(x) = -1$. (g) $1 \neq -1$: $\lim_{x\to 1} f(x)$ does not exist. Value: $f(1) = 2$, the filled dot above the gap. It is a perfectly good value, and it is irrelevant for all three limits — the most common mistake is to answer (g) with "2".

At 3 (h) From the left the graph climbs higher and higher along the dashed asymptote: $\lim_{x\to 3^-} f(x) = +\infty$. (i) From the right the graph comes up from far below: $\lim_{x\to 3^+} f(x) = -\infty$. (j) The two sides go opposite ways, so $\lim_{x\to 3} f(x)$ does not exist — not even as an infinite limit. (If both sides had gone to $+\infty$ we would write $\lim_{x\to 3} f(x) = +\infty$.) Value: $f(3)$ is undefined — there is no dot on the line $x = 3$; a vertical asymptote is not part of the graph.

Problem 2 *Using the limit laws.*

Suppose that $\lim_{x\to 2} f(x) = 3$, $\lim_{x\to 2} g(x) = -2$ and $\lim_{x\to 2} h(x) = 0$. Evaluate the following limits as $x \to 2$, naming the law you use at each step — or explain why the laws do not give the answer.

(a) $\big[f(x)+3g(x)\big]$

(b) $f(x)\,g(x)-h(x)$

(c) $\dfrac{f(x)}{g(x)}$

(d) $\dfrac{g(x)}{h(x)}$

(e) $\sqrt{f(x)+1}\,\big[g(x)\big]^2$

(f) $\dfrac{h(x)}{f(x)}$

Solution. Each law may be used only when the limits it needs exist — and the quotient law also needs the limit of the denominator to be non-zero.

(a) Sum law, then constant-multiple law: $\lim\big[f(x)+3g(x)\big] = \lim f(x)+3\lim g(x) = 3+3\cdot(-2) = -3$.

(b) Difference law, then product law: $\lim\big[f(x)g(x)-h(x)\big] = \lim f(x)\cdot\lim g(x) - \lim h(x) = 3\cdot(-2)-0 = -6$.

(c) Quotient law: the denominator's limit is $-2\neq 0$, so the law applies: $\lim\dfrac{f(x)}{g(x)} = \dfrac{3}{-2} = -\dfrac{3}{2}$.

(d) The quotient law does *not* apply, because $\lim h(x)=0$. And no other law rescues it: near 2 the numerator is close to -2 while the denominator is close to 0, so $|g(x)/h(x)|$ becomes arbitrarily large — there is no real limit. Whether the quotient tends to $+\infty$, to $-\infty$, or to neither depends on the sign of $h(x)$ on each side, which we are not told.

(e) Root law (allowed because $\lim[f(x)+1] = 4 > 0$): $\lim\sqrt{f(x)+1} = \sqrt{3+1} = 2$. Power law: $\lim[g(x)]^2 = (-2)^2 = 4$. Product law: $2\cdot 4 = 8$.

(f) Quotient law, denominator's limit $3\neq 0$: $\lim\dfrac{h(x)}{f(x)} = \dfrac{0}{3} = 0$. A zero *in the numerator* is harmless; only a zero in the denominator causes trouble.

Problem 3 $0/0$: *factor and cancel.*

In each limit, substitution gives $0/0$. Factor, cancel the common factor — and say in one sentence why you may cancel — then substitute.

(a) $\displaystyle\lim_{x\to 4}\frac{x^2-16}{x^2-5x+4}$

(b) $\displaystyle\lim_{x\to -2}\frac{x^2+5x+6}{x^2-4}$

(c) $\displaystyle\lim_{x\to 3}\frac{2x^2-5x-3}{x-3}$

(d) $\displaystyle\lim_{h\to 0}\frac{(1+2h)^2-1}{h}$

(e) $\displaystyle\lim_{t\to 1}\frac{t^3-1}{t^2-1}$

Solution. The pattern is the same every time: (1) substitute — you get $0/0$, which tells you the numerator and the denominator share a factor that vanishes at a; (2) factor both; (3) divide the numerator and the denominator by that common factor; (4) substitute in what is left. **Say why step (3) is legitimate, every time:** one may divide by a factor *only where that factor is not zero*, i.e. only for $x\neq a$ — and the limit as $x\to a$ considers only $x\neq a$ (the definition says $0<|x-a|<\delta$). The two functions, before and after the cancellation, are different at a and equal everywhere else, so they have the same limit at a. The limit sign stays in front of every expression until the very last step, where the limit of the simplified expression is found by substitution.

(a) At $x=4$: numerator $16-16=0$, denominator $16-20+4=0$. Factor: $x^2-16=(x-4)(x+4)$; $x^2-5x+4=(x-1)(x-4)$ (its roots are 1 and 4). Divide by $x-4$ — allowed only where $x-4\neq 0$, i.e. for $x\neq 4$, and the limit at 4 uses only such x:

$$\lim_{x\to 4}\frac{x^2-16}{x^2-5x+4} = \lim_{x\to 4}\frac{(x-4)(x+4)}{(x-1)(x-4)} = \lim_{x\to 4}\frac{x+4}{x-1} = \frac{4+4}{4-1} = \frac{8}{3}.$$

(b) At $x=-2$: $4-10+6=0$ and $4-4=0$. Factor: $x^2+5x+6=(x+2)(x+3)$; $x^2-4=(x-2)(x+2)$. Divide by $x+2$ — only where $x+2\neq 0$, i.e. for $x\neq -2$:

$$\lim_{x\to -2}\frac{x^2+5x+6}{x^2-4} = \lim_{x\to -2}\frac{(x+2)(x+3)}{(x-2)(x+2)} = \lim_{x\to -2}\frac{x+3}{x-2} = \frac{-2+3}{-2-2} = -\frac{1}{4}.$$

(c) At $x=3$: $18-15-3=0$ and $3-3=0$. Since $x=3$ is a root of $2x^2-5x-3$, $(x-3)$ is a factor; dividing (or guessing the second factor) gives $2x^2-5x-3=(x-3)(2x+1)$ — check:

$(x-3)(2x+1) = 2x^2+x-6x-3 = 2x^2-5x-3$. Divide by $x-3$ — only where $x-3 \neq 0$, i.e. for $x \neq 3$:

$$\lim_{x\to 3} \frac{2x^2-5x-3}{x-3} = \lim_{x\to 3} \frac{(x-3)(2x+1)}{x-3} = \lim_{x\to 3}(2x+1) = 2\cdot 3+1 = 7.$$

(d) At $h=0$: $(1+0)^2-1=0$ over 0. Expand: $(1+2h)^2 = 1+4h+4h^2$, so $(1+2h)^2-1 = 4h+4h^2 = h(4+4h)$. Divide by h — only where $h \neq 0$, and the limit as $h \to 0$ uses only such h:

$$\lim_{h\to 0} \frac{(1+2h)^2-1}{h} = \lim_{h\to 0} \frac{4h+4h^2}{h} = \lim_{h\to 0} \frac{h(4+4h)}{h} = \lim_{h\to 0}(4+4h) = 4.$$

(e) At $t=1$: 0/0. Factor: $t^3-1 = (t-1)(t^2+t+1)$ and $t^2-1 = (t-1)(t+1)$. Divide by $t-1$ — only where $t-1 \neq 0$, i.e. for $t \neq 1$:

$$\lim_{t\to 1} \frac{t^3-1}{t^2-1} = \lim_{t\to 1} \frac{(t-1)(t^2+t+1)}{(t-1)(t+1)} = \lim_{t\to 1} \frac{t^2+t+1}{t+1} = \frac{1+1+1}{1+1} = \frac{3}{2}.$$

Problem 4 0/0*: multiply by the conjugate.*

Again 0/0 every time. Multiply top and bottom by the conjugate of the expression with the root, simplify, cancel, substitute.

(a) $\lim_{x\to 9} \dfrac{\sqrt{x}-3}{x-9}$

(b) $\lim_{h\to 0} \dfrac{\sqrt{4+h}-2}{h}$

(c) $\lim_{x\to 2} \dfrac{\sqrt{5x-1}-3}{x-2}$

(d) $\lim_{x\to 1} \dfrac{x-1}{\sqrt{x+3}-2}$

(e) $\lim_{x\to 0} \dfrac{\sqrt{2+x}-\sqrt{2-x}}{x}$

Solution. The conjugate of $\sqrt{A}-\sqrt{B}$ is $\sqrt{A}+\sqrt{B}$, and $(\sqrt{A}-\sqrt{B})(\sqrt{A}+\sqrt{B}) = A-B$: multiplying top and bottom by the conjugate removes the roots from the difference and makes the factor that caused the 0/0 visible. (Multiplying by $\frac{\text{conjugate}}{\text{conjugate}}$ is itself legitimate because the conjugate is a *sum* of positive quantities near the point, so it is never zero there.) Do not expand the product that contains the root — leave it as it is and divide by the common factor, saying each time for which x that division is allowed.

(a) At $x=9$: $\sqrt{9}-3=0$ over $9-9=0$. Multiply top and bottom by $\sqrt{x}+3$, then divide by $x-9$ — allowed only where $x-9 \neq 0$, i.e. for $x \neq 9$:

$$\lim_{x\to 9} \frac{\sqrt{x}-3}{x-9} = \lim_{x\to 9} \frac{(\sqrt{x}-3)(\sqrt{x}+3)}{(x-9)(\sqrt{x}+3)} = \lim_{x\to 9} \frac{x-9}{(x-9)(\sqrt{x}+3)} = \lim_{x\to 9} \frac{1}{\sqrt{x}+3} = \frac{1}{3+3} = \frac{1}{6}.$$

(Alternatively, factor the denominator: $x-9 = (\sqrt{x}-3)(\sqrt{x}+3)$.)

(b) At $h=0$: $\sqrt{4}-2=0$ over 0. Multiply by $\sqrt{4+h}+2$, then divide by h — allowed only for $h \neq 0$:

$$\lim_{h\to 0} \frac{\sqrt{4+h}-2}{h} = \lim_{h\to 0} \frac{(4+h)-4}{h\,(\sqrt{4+h}+2)} = \lim_{h\to 0} \frac{h}{h\,(\sqrt{4+h}+2)} = \lim_{h\to 0} \frac{1}{\sqrt{4+h}+2} = \frac{1}{2+2} = \frac{1}{4}.$$

(c) At $x=2$: $\sqrt{9}-3=0$ over 0. Multiply by $\sqrt{5x-1}+3$; the numerator becomes $(5x-1)-9 = 5x-10 = 5(x-2)$. Divide by $x-2$ — only for $x \neq 2$:

$$\lim_{x\to 2} \frac{\sqrt{5x-1}-3}{x-2} = \lim_{x\to 2} \frac{5(x-2)}{(x-2)(\sqrt{5x-1}+3)} = \lim_{x\to 2} \frac{5}{\sqrt{5x-1}+3} = \frac{5}{3+3} = \frac{5}{6}.$$

(d) At $x=1$: 0 over $\sqrt{4}-2=0$. The root is in the denominator, so the conjugate $\sqrt{x+3}+2$ goes there; the denominator becomes $(x+3)-4 = x-1$. Divide by $x-1$ — only for $x \neq 1$:

$$\lim_{x\to 1} \frac{x-1}{\sqrt{x+3}-2} = \lim_{x\to 1} \frac{(x-1)(\sqrt{x+3}+2)}{x-1} = \lim_{x\to 1}\left(\sqrt{x+3}+2\right) = 2+2 = 4.$$

(e) At $x = 0$: $\sqrt{2} - \sqrt{2} = 0$ over 0. Two roots, one conjugate $\sqrt{2+x} + \sqrt{2-x}$; the numerator becomes $(2+x) - (2-x) = 2x$. Divide by x — only for $x \neq 0$:

$$\lim_{x\to 0} \frac{\sqrt{2+x} - \sqrt{2-x}}{x} = \lim_{x\to 0} \frac{2x}{x\,(\sqrt{2+x} + \sqrt{2-x})} = \lim_{x\to 0} \frac{2}{\sqrt{2+x} + \sqrt{2-x}} = \frac{2}{2\sqrt{2}} = \frac{\sqrt{2}}{2}.$$

Problem 5 $0/0$ *(and $\infty - \infty$): combine the fractions first.*

Bring the fractions to one denominator, simplify, cancel, substitute. In (d) and (e) substitution gives not $0/0$ but "$\infty - \infty$", which decides nothing either.

(a) $\displaystyle\lim_{x\to 5} \frac{\frac{1}{x} - \frac{1}{5}}{x-5}$

(b) $\displaystyle\lim_{h\to 0} \frac{\frac{1}{4+h} - \frac{1}{4}}{h}$

(c) $\displaystyle\lim_{x\to -3} \frac{\frac{1}{3} + \frac{1}{x}}{x+3}$

(d) $\displaystyle\lim_{t\to 0}\Big(\frac{1}{t} - \frac{2}{t^2+2t}\Big)$

(e) $\displaystyle\lim_{x\to 2}\Big(\frac{1}{x-2} - \frac{4}{x^2-4}\Big)$

Solution. When the numerator is itself a difference of fractions, combine it into one fraction *first*; then divide — and, as always, say for which x the division by the common factor is allowed: only where that factor is not zero, which is exactly the set of x the limit looks at. In (d) and (e) substitution does not even give $0/0$ — each term separately blows up ("$\infty - \infty$"), which decides nothing; combining the fractions cancels the blow-up.

(a) At $x = 5$: $\frac{1}{5} - \frac{1}{5} = 0$ over 0. Combine the numerator: $\dfrac{1}{x} - \dfrac{1}{5} = \dfrac{5-x}{5x}$, and note $5 - x = -(x-5)$. Divide by $x - 5$ — only where $x - 5 \neq 0$, i.e. for $x \neq 5$:

$$\lim_{x\to 5} \frac{\frac{1}{x} - \frac{1}{5}}{x-5} = \lim_{x\to 5} \frac{5-x}{5x\,(x-5)} = \lim_{x\to 5} \frac{-(x-5)}{5x\,(x-5)} = \lim_{x\to 5}\Big(-\frac{1}{5x}\Big) = -\frac{1}{25}.$$

(b) At $h = 0$: $\frac{1}{4} - \frac{1}{4} = 0$ over 0. Combine: $\dfrac{1}{4+h} - \dfrac{1}{4} = \dfrac{4-(4+h)}{4(4+h)} = \dfrac{-h}{4(4+h)}$. Then divide by h — allowed only for $h \neq 0$:

$$\lim_{h\to 0} \frac{\frac{1}{4+h} - \frac{1}{4}}{h} = \lim_{h\to 0} \frac{-h}{4(4+h)\,h} = \lim_{h\to 0} \frac{-1}{4(4+h)} = -\frac{1}{16}.$$

(c) At $x = -3$: $\frac{1}{3} + \frac{1}{-3} = 0$ over 0. Combine: $\dfrac{1}{3} + \dfrac{1}{x} = \dfrac{x+3}{3x}$. Then divide by $x + 3$ — allowed only where $x + 3 \neq 0$, i.e. for $x \neq -3$:

$$\lim_{x\to -3} \frac{\frac{1}{3} + \frac{1}{x}}{x+3} = \lim_{x\to -3} \frac{x+3}{3x\,(x+3)} = \lim_{x\to -3} \frac{1}{3x} = \frac{1}{-9} = -\frac{1}{9}.$$

(d) As $t \to 0$ both $\frac{1}{t}$ and $\frac{2}{t^2+2t}$ are unbounded: "$\infty - \infty$". Factor $t^2 + 2t = t(t+2)$ and use the common denominator $t(t+2)$; then divide by t — only for $t \neq 0$:

$$\lim_{t\to 0}\Big(\frac{1}{t} - \frac{2}{t^2+2t}\Big) = \lim_{t\to 0} \frac{(t+2)-2}{t(t+2)} = \lim_{t\to 0} \frac{t}{t(t+2)} = \lim_{t\to 0} \frac{1}{t+2} = \frac{1}{2}.$$

(e) As $x \to 2$ both terms are unbounded. Factor $x^2 - 4 = (x-2)(x+2)$ and use the common denominator $(x-2)(x+2)$; then divide by $x - 2$ — only for $x \neq 2$:

$$\lim_{x\to 2}\Big(\frac{1}{x-2} - \frac{4}{x^2-4}\Big) = \lim_{x\to 2} \frac{(x+2)-4}{(x-2)(x+2)} = \lim_{x\to 2} \frac{x-2}{(x-2)(x+2)} = \lim_{x\to 2} \frac{1}{x+2} = \frac{1}{4}.$$

Problem 6 *One-sided limits and functions defined in pieces.*

(a) Let

$$f(x) = \begin{cases} x^2 + 1, & x < 1, \\ 3 - x, & 1 \le x < 3, \\ \sqrt{x-3}, & x \ge 3. \end{cases}$$

Find the one-sided limits of f at 1 and at 3; decide whether $\lim_{x\to 1} f(x)$ and $\lim_{x\to 3} f(x)$ exist, and compare them with $f(1)$ and $f(3)$. Sketch the graph.

(b) Find $\lim\limits_{x\to 2^-} \dfrac{|x-2|}{x-2}$, $\lim\limits_{x\to 2^+} \dfrac{|x-2|}{x-2}$ and decide whether $\lim\limits_{x\to 2} \dfrac{|x-2|}{x-2}$ exists.

(c) Find the one-sided limits at 3 and decide whether the two-sided limit exists, for (i) $\dfrac{x^2-9}{|x-3|}$ and (ii) $\dfrac{x^2-6x+9}{|x-3|}$. Why does one of the limits exist and the other not?

(d) Find $\lim\limits_{x\to 0}\left(\dfrac{1}{|x|} - \dfrac{1}{x}\right)$. (Careful: treat the two sides separately.)

Solution.

(a) *At* 1. For $x < 1$ the formula is $x^2 + 1$, so $\lim_{x\to 1^-} f(x) = 1^2 + 1 = 2$. For $x > 1$ (and $x < 3$) the formula is $3 - x$, so $\lim_{x\to 1^+} f(x) = 3 - 1 = 2$. The one-sided limits agree, so $\lim_{x\to 1} f(x) = 2$. The value: $x = 1$ falls in the middle piece ($1 \le x < 3$), so $f(1) = 3 - 1 = 2$ — limit and value agree; the formula changes at 1 but the graph does not jump.
At 3. For $x < 3$: $\lim_{x\to 3^-} f(x) = 3 - 3 = 0$. For $x \ge 3$: $\lim_{x\to 3^+} f(x) = \sqrt{3-3} = 0$. So $\lim_{x\to 3} f(x) = 0$, and $f(3) = \sqrt{0} = 0$.
Sketch: the arc of the parabola $y = x^2 + 1$ for $x < 1$, rising to the point $(1, 2)$; the segment of $y = 3 - x$ from $(1, 2)$ down to $(3, 0)$; then the half-parabola $y = \sqrt{x-3}$ starting at $(3, 0)$ and rising slowly. The three pieces join without gaps.

(b) By definition $|x-2| = x - 2$ when $x > 2$ and $|x-2| = -(x-2)$ when $x < 2$. So for $x > 2$ the quotient is $\dfrac{x-2}{x-2} = 1$, giving $\lim_{x\to 2^+} = 1$; for $x < 2$ it is $\dfrac{-(x-2)}{x-2} = -1$, giving $\lim_{x\to 2^-} = -1$. The one-sided limits differ, so $\lim_{x\to 2} \dfrac{|x-2|}{x-2}$ does not exist.

(c) (i) Factor: $x^2 - 9 = (x-3)(x+3)$; dividing by $x - 3$ is allowed only for $x \ne 3$, which is all the limit looks at. For $x > 3$, $|x-3| = x-3$ and $\dfrac{(x-3)(x+3)}{x-3} = x+3 \to 6$. For $x < 3$, $|x-3| = -(x-3)$ and $\dfrac{(x-3)(x+3)}{-(x-3)} = -(x+3) \to -6$. Right limit 6, left limit -6: no two-sided limit.

(ii) $x^2 - 6x + 9 = (x-3)^2$, and $(x-3)^2 = |x-3|^2$. So for $x \ne 3$ the quotient is $\dfrac{|x-3|^2}{|x-3|} = |x-3|$, which tends to 0 from both sides: $\lim_{x\to 3} = 0$.
Why the difference? In (i) the quotient contains the factor $\dfrac{x-3}{|x-3|}$, which is $+1$ on the right of 3 and -1 on the left — the sign flips, so the two sides cannot agree. In (ii) the numerator is a square, $(x-3)^2 \ge 0$, and there is no sign left to flip.

(d) For $x > 0$: $|x| = x$, so $\dfrac{1}{|x|} - \dfrac{1}{x} = \dfrac{1}{x} - \dfrac{1}{x} = 0$; the right-hand limit is 0. For $x < 0$: $|x| = -x$, so $\dfrac{1}{|x|} - \dfrac{1}{x} = -\dfrac{1}{x} - \dfrac{1}{x} = -\dfrac{2}{x}$; as $x \to 0^-$, x is a small negative number, so $-\dfrac{2}{x}$ is a large positive number: the left-hand limit is $+\infty$. The two sides disagree: the limit does not exist.

Problem 7 *Infinite limits and vertical asymptotes.*

Determine these limits. Find all vertical asymptotes of these functions.

(a) $\lim\limits_{x\to 3} \dfrac{x+2}{x-3}$

(b) $\lim\limits_{x\to 0} \dfrac{x-2}{x^2(x+3)}$

(c) $\lim\limits_{x\to 2} \dfrac{x+2}{x^2-4}$

(d) $\lim\limits_{x\to 1} \dfrac{x^2-4}{x^2-3x+2}$

Solution. For a quotient whose denominator tends to 0 while the numerator tends to a non-zero number, the quotient is unbounded; its *sign* on each side decides between $+\infty$ and $-\infty$. To find the vertical asymptotes, check *every* zero of the denominator: if the function is unbounded there, it is an asymptote; if the zero cancels against the numerator and the function has a finite limit there, it is a hole, not an asymptote.

(a) The numerator tends to $5 > 0$. From the right ($x > 3$) the denominator $x - 3$ is a small positive number, so the quotient is large and positive: $\lim_{x\to 3^+} = +\infty$. From the left ($x < 3$) the denominator is a small negative number: $\lim_{x\to 3^-} = -\infty$. The two-sided limit does not exist (the two sides go opposite ways). *Asymptotes:* the denominator vanishes only at 3, and there the function is unbounded — exactly one vertical asymptote, $x = 3$.

(b) The numerator tends to $-2 < 0$. The denominator $x^2(x+3)$: x^2 is a small *positive* number on both sides of 0, and $x + 3 \to 3 > 0$; so the denominator is small and positive on both sides. Negative over small positive: $-\infty$ on both sides, hence $\lim_{x\to 0} = -\infty$. *Asymptotes:* the denominator vanishes at 0 and at -3. At -3 the numerator tends to $-5 \neq 0$, so the function is unbounded there as well (write it as $\frac{x-2}{x^2} \cdot \frac{1}{x+3}$: the first factor tends to $-\frac{5}{9}$, the second to $\pm\infty$). Two vertical asymptotes: $x = 0$ and $x = -3$.

(c) Factor: $x^2 - 4 = (x-2)(x+2)$ and divide by $x + 2$ — allowed only where $x + 2 \neq 0$, i.e. for $x \neq -2$; so for $x \neq -2$ the function equals $\dfrac{x+2}{(x-2)(x+2)} = \dfrac{1}{x-2}$. At 2: $\dfrac{1}{x-2} \to +\infty$ from the right and $-\infty$ from the left; the two-sided limit does not exist. *Asymptotes:* the denominator vanishes at 2 and at -2. At 2 the function is unbounded: asymptote. At -2 the function equals $\dfrac{1}{x-2} \to \dfrac{1}{-4}$ — a finite limit, so $x = -2$ is a *hole* in the graph, not an asymptote. One vertical asymptote, $x = 2$.

(d) Factor: $x^2 - 4 = (x-2)(x+2)$ and $x^2 - 3x + 2 = (x-1)(x-2)$; dividing by $x - 2$ is allowed only where $x - 2 \neq 0$, i.e. for $x \neq 2$, and there the function equals $\dfrac{x+2}{x-1}$. At 1: the numerator tends to $3 > 0$; the denominator $x - 1$ is negative for $x < 1$ and positive for $x > 1$; so $\lim_{x\to 1^-} = -\infty$, $\lim_{x\to 1^+} = +\infty$, and the two-sided limit does not exist. *Asymptotes:* the denominator vanishes at 1 and at 2. At 1: unbounded, asymptote. At 2: the function equals $\dfrac{x+2}{x-1} \to \dfrac{4}{1} = 4$, a hole. One vertical asymptote, $x = 1$. (The rule of thumb: a zero of the denominator is where to *look*; whether it is an asymptote or a hole depends on whether the numerator vanishes there too.)

Problem 8 *The Squeeze Theorem.*

(a) Show that $\lim\limits_{x\to 0} x^2 \cos\dfrac{1}{x^2} = 0$.

(b) If $3x - 2 \le f(x) \le x^2 - x + 2$ for all $x \ge 0$, find $\lim\limits_{x\to 2} f(x)$.

(c) Show that $\lim\limits_{x\to 0^+} \sqrt{x}\left(1 + \sin^2\dfrac{2\pi}{x}\right) = 0$.

(d) Why does the Squeeze Theorem *not* show that $\lim\limits_{x\to 0} \cos\dfrac{1}{x}$ exists, although $-1 \le \cos\frac{1}{x} \le 1$?

Solution. The Squeeze Theorem: if $g(x) \le f(x) \le h(x)$ for all x near a (except possibly at a) and $\lim_{x\to a} g(x) = \lim_{x\to a} h(x) = L$, then $\lim_{x\to a} f(x) = L$. To use it you must *write down the two bounds* and check that they have the *same* limit.

(a) The cosine of anything lies between -1 and 1: $-1 \le \cos\dfrac{1}{x^2} \le 1$ for $x \neq 0$. Multiply through by x^2, which is positive (so the inequalities keep their direction): $-x^2 \le x^2\cos\dfrac{1}{x^2} \le x^2$. Both bounds tend to 0 as $x \to 0$. By the Squeeze Theorem, $\lim_{x\to 0} x^2\cos\dfrac{1}{x^2} = 0$. (The function is not defined at 0; the theorem does not care.)

(b) We are told $3x - 2 \le f(x) \le x^2 - x + 2$. The two bounds are polynomials, so their limits at 2 are their values: $3\cdot 2 - 2 = 4$ and $2^2 - 2 + 2 = 4$. Same limit 4 on both sides, so by the Squeeze Theorem $\lim_{x\to 2} f(x) = 4$. (Consistency check: the difference of the bounds is $x^2 - 4x + 4 = (x-2)^2 \ge 0$, so the lower bound really is below the upper one, and they touch exactly at $x = 2$ — which is why the squeeze works there.)

(c) $0 \le \sin^2\dfrac{2\pi}{x} \le 1$, so $1 \le 1+\sin^2\dfrac{2\pi}{x} \le 2$. Multiply by $\sqrt{x} > 0$ (for $x > 0$): $\sqrt{x} \le \sqrt{x}\left(1+\sin^2\dfrac{2\pi}{x}\right) \le 2\sqrt{x}$. As $x \to 0^+$ both $\sqrt{x}$ and $2\sqrt{x}$ tend to 0; by the (one-sided) Squeeze Theorem the limit is 0.

(d) The theorem needs the two bounds to have the *same* limit. Here the bounds are the constants -1 and 1, whose limits are -1 and 1 — different — so the theorem says nothing at all. And indeed $\cos\dfrac{1}{x}$ has no limit at 0: at $x = \dfrac{1}{2\pi n}$ its value is 1, at $x = \dfrac{1}{(2n+1)\pi}$ its value is -1, and both kinds of points come arbitrarily close to 0. A function that keeps taking the values 1 and -1 near 0 cannot approach a single number.

Part II · The precise definition of a limit (Section 1.7) · 50 minutes

Warm-up II · True or false?

Throughout, $\lim_{x\to a} f(x) = L$ is meant in the precise sense: for every $\varepsilon > 0$ there is $\delta > 0$ such that $0 < |x - a| < \delta$ implies $|f(x) - L| < \varepsilon$.

1. If $|f(x) - L| < 0.01$ whenever $0 < |x - a| < 0.5$, then $\lim_{x\to a} f(x) = L$.
2. In the definition, δ is allowed to depend on ε.
3. If some δ works for a given ε, then every smaller δ works for that ε too.
4. If some δ works for a given ε, then the same δ works for every smaller ε.
5. For $\lim_{x\to 3}(2x - 1) = 5$ and $\varepsilon = 0.1$, the choice $\delta = 0.05$ works.
6. For the same limit and $\varepsilon = 0.1$, the choice $\delta = 0.1$ also works.
7. The definition requires $f(a) = L$.
8. "$\lim_{x\to a} f(x) = \infty$" means that $f(a) = \infty$.

Solution.

1 False. The definition demands the inequality for *every* $\varepsilon > 0$; one value $\varepsilon = 0.01$ proves nothing. Example: if $f(x) = L + 0.005$ for all $x \neq a$, then $|f(x) - L| = 0.005 < 0.01$ for every x, yet $\lim_{x\to a} f(x) = L + 0.005 \neq L$.

2 True. δ is chosen *after* ε, as a response to it, and normally depends on it — in the lecture game with $f(x) = 2x - 1$ the response was $\delta = \varepsilon/2$.

3 True. Suppose δ works and $\delta' < \delta$. If $0 < |x - a| < \delta'$ then also $0 < |x - a| < \delta$, so $|f(x) - L| < \varepsilon$ holds already. A smaller δ allows fewer x, and every one of them was already fine. *Smaller is always safe.*

4 False. A smaller ε is a *harder* challenge and usually needs a smaller δ. For $f(x) = 2x - 1$ at 3, $\delta = 0.05$ works for $\varepsilon = 0.1$ (see item 5), but for $\varepsilon = 0.01$ it fails: $x = 3.04$ satisfies $|x - 3| < 0.05$, yet $|f(x) - 5| = 2\cdot 0.04 = 0.08 > 0.01$.

5 True. $|(2x - 1) - 5| = |2x - 6| = 2|x - 3|$; if $|x - 3| < 0.05$ then $2|x - 3| < 0.1$.

6 False. Take $x = 3.08$: then $|x-3| = 0.08 < 0.1$, but $|(2x-1)-5| = 2 \cdot 0.08 = 0.16$, which is not < 0.1. A δ that lets one bad x through does not work.

7 False. The hypothesis is $0 < |x-a| < \delta$; the "$0 <$" excludes $x = a$, so $f(a)$ is never tested. It may be any number, or undefined — $\dfrac{x^2-4}{x-2}$ has limit 4 at 2 and no value there.

8 False. ∞ is not a number, so "$f(a) = \infty$" is meaningless. "$\lim_{x\to a} f(x) = \infty$" is a statement about the values of f for x *near* a: for every M there is $\delta > 0$ such that $f(x) > M$ whenever $0 < |x-a| < \delta$. The value $f(a)$ plays no role and may even be defined (Warm-up I, item 9).

Problem 9 *Linear ε–δ proofs.*

Prove each statement from the definition. Write the scratch work (finding δ backwards) on the left and the proof (forward) on the right, separated by a line.

(a) $\lim\limits_{x\to 2}(5x-3) = 7$

(b) $\lim\limits_{x\to -1}(2-3x) = 5$

(c) $\lim\limits_{x\to 4}\left(\dfrac{x}{2}+1\right) = 3$

(d) $\lim\limits_{x\to -2}(4x+9) = 1$

(e) $\lim\limits_{x\to 1}(7-2x) = 5$

(f) $\lim\limits_{x\to 3}\dfrac{x^2-x-6}{x-3} = 5$

Then, for (a): which δ does your proof give when $\varepsilon = 0.01$? And is $\delta = 0.01$ itself good enough for $\varepsilon = 0.01$?

Solution. Every proof has the same shape. *Scratch work* (backwards): compute $|f(x)-L|$, rewrite it as a multiple of $|x-a|$, and see how small $|x-a|$ must be for $|f(x)-L| < \varepsilon$; that tells you δ. *Proof* (forward): "Let $\varepsilon > 0$. Choose $\delta = \ldots$. If $0 < |x-a| < \delta$, then $|f(x)-L| = \cdots < \varepsilon$." Four things must be present: "let $\varepsilon > 0$" first; a δ that depends on ε; the chain of (in)equalities ending in ε; a concluding sentence.

(a) Scratch: $|(5x-3)-7| = |5x-10| = 5|x-2|$. We want $5|x-2| < \varepsilon$, i.e. $|x-2| < \varepsilon/5$; so $\delta = \varepsilon/5$ should work. Proof: Let $\varepsilon > 0$. Choose $\delta = \varepsilon/5$. If $0 < |x-2| < \delta$, then $|(5x-3)-7| = 5|x-2| < 5\delta = 5 \cdot \dfrac{\varepsilon}{5} = \varepsilon$. Hence $\lim_{x\to 2}(5x-3) = 7$. □

Follow-up. For $\varepsilon = 0.01$ the proof gives $\delta = 0.01/5 = 0.002$. Is $\delta = 0.01$ itself good enough? No: $x = 2.008$ satisfies $|x-2| = 0.008 < 0.01$, but $|(5x-3)-7| = 5 \cdot 0.008 = 0.04 > 0.01$. The slope 5 magnifies distances five times, so δ must be five times smaller than ε.

(b) Scratch: $|(2-3x)-5| = |-3x-3| = |-3(x+1)| = 3|x+1| = 3|x-(-1)|$. We want $3|x+1| < \varepsilon$, i.e. $|x+1| < \varepsilon/3$. Proof: Let $\varepsilon > 0$; choose $\delta = \varepsilon/3$. If $0 < |x-(-1)| < \delta$, then $|(2-3x)-5| = 3|x+1| < 3\delta = \varepsilon$. □

(c) Scratch: $\left|\left(\frac{x}{2}+1\right)-3\right| = \left|\frac{x}{2}-2\right| = \frac{1}{2}|x-4|$. We want $\frac{1}{2}|x-4| < \varepsilon$, i.e. $|x-4| < 2\varepsilon$. Proof: Let $\varepsilon > 0$; choose $\delta = 2\varepsilon$. If $0 < |x-4| < \delta$, then $\left|\frac{x}{2}+1-3\right| = \frac{1}{2}|x-4| < \frac{1}{2} \cdot 2\varepsilon = \varepsilon$. □ (Here δ is *larger* than ε: a function with slope $\frac{1}{2}$ shrinks distances, so it is easy to control.)

(d) Scratch: $|(4x+9)-1| = |4x+8| = 4|x+2| = 4|x-(-2)|$; we want $|x+2| < \varepsilon/4$. Proof: Let $\varepsilon > 0$; choose $\delta = \varepsilon/4$. If $0 < |x+2| < \delta$, then $|(4x+9)-1| = 4|x+2| < 4\delta = \varepsilon$. □

(e) Scratch: $|(7-2x)-5| = |2-2x| = |2(1-x)| = 2|x-1|$ (because $|1-x| = |x-1|$); we want $|x-1| < \varepsilon/2$. Proof: Let $\varepsilon > 0$; choose $\delta = \varepsilon/2$. If $0 < |x-1| < \delta$, then $|(7-2x)-5| = 2|x-1| < 2\delta = \varepsilon$. □ (A negative slope changes nothing: the absolute value takes care of the sign.)

(f) Scratch: for $x \neq 3$, $\dfrac{x^2-x-6}{x-3} = \dfrac{(x-3)(x+2)}{x-3} = x+2$, so $\left|\dfrac{x^2-x-6}{x-3}-5\right| = |(x+2)-5| = |x-3|$; we want $|x-3| < \varepsilon$, so $\delta = \varepsilon$. Proof: Let $\varepsilon > 0$; choose $\delta = \varepsilon$. If $0 < |x-3| < \delta$, then $x \neq 3$, so the function equals $x+2$ and $\left|\dfrac{x^2-x-6}{x-3}-5\right| = |x-3| < \delta = \varepsilon$. □ Notice where "$0 < |x-3|$" was used: it guarantees $x \neq 3$, which is exactly what makes the cancellation legitimate — the function is not even defined at 3, and the proof never needs it to be.

Problem 10 *When the factor is not a constant: the* min *trick.*

(a) Prove that $\lim_{x\to 2} x^2 = 4$. (Hint: $|x^2-4| = |x-2|\,|x+2|$. First agree that $|x-2|<1$, and find how large $|x+2|$ can then be.)

(b) Prove that $\lim_{x\to 3}(x^2-2x) = 3$.

(c) Prove that $\lim_{x\to -1}(x^2+3x) = -2$.

(d) Prove that $\lim_{x\to 1} x^3 = 1$. (Hint: $x^3-1 = (x-1)(x^2+x+1)$.)

(e) Prove that $\lim_{x\to 1}\dfrac{1}{x} = 1$. (Hint: $\left|\dfrac{1}{x}-1\right| = \dfrac{|x-1|}{|x|}$; keep x away from 0.)

Solution. For a non-linear function, $|f(x)-L|$ is $|x-a|$ times a factor that is *not constant*. The trick: first agree that x stays within a fixed distance of a (say $|x-a|<1$); on that interval the extra factor is bounded by some number K; then $|f(x)-L| < K|x-a|$, which is $<\varepsilon$ as soon as $|x-a|<\varepsilon/K$. Both requirements must hold, so $\delta = \min(1, \varepsilon/K)$ — the smaller of the two numbers satisfies both conditions.

(a) Scratch: $|x^2-4| = |x-2|\,|x+2|$. The factor $|x+2|$ is not constant — tame it: if $|x-2|<1$ then $1<x<3$, so $3<x+2<5$ and $|x+2|<5$. Then $|x^2-4|<5|x-2|$, which is $<\varepsilon$ when $|x-2|<\varepsilon/5$. Proof: Let $\varepsilon>0$. Choose $\delta = \min(1,\varepsilon/5)$. If $0<|x-2|<\delta$, then $|x-2|<1$, so $|x+2|<5$; and $|x-2|<\varepsilon/5$; therefore $|x^2-4| = |x-2|\,|x+2| < \dfrac{\varepsilon}{5}\cdot 5 = \varepsilon$. □

(b) Scratch: $|(x^2-2x)-3| = |x^2-2x-3| = |(x-3)(x+1)| = |x-3|\,|x+1|$. If $|x-3|<1$ then $2<x<4$, so $3<x+1<5$ and $|x+1|<5$. Proof: Let $\varepsilon>0$; choose $\delta=\min(1,\varepsilon/5)$. If $0<|x-3|<\delta$, then $|x+1|<5$ and $|(x^2-2x)-3| = |x-3|\,|x+1| < \dfrac{\varepsilon}{5}\cdot 5 = \varepsilon$. □

(c) Scratch: $|(x^2+3x)-(-2)| = |x^2+3x+2| = |(x+1)(x+2)| = |x+1|\,|x+2|$. If $|x+1|<1$ then $-2<x<0$, so $0<x+2<2$ and $|x+2|<2$. Proof: Let $\varepsilon>0$; choose $\delta=\min(1,\varepsilon/2)$. If $0<|x+1|<\delta$, then $|x+2|<2$ and $|(x^2+3x)+2| = |x+1|\,|x+2| < \dfrac{\varepsilon}{2}\cdot 2 = \varepsilon$. □

(d) Scratch: $|x^3-1| = |x-1|\,|x^2+x+1|$. If $|x-1|<1$ then $0<x<2$, so $x^2<4$, $x<2$, and $0<x^2+x+1<4+2+1=7$. Proof: Let $\varepsilon>0$; choose $\delta=\min(1,\varepsilon/7)$. If $0<|x-1|<\delta$, then $|x^2+x+1|<7$ and $|x^3-1| = |x-1|\,|x^2+x+1| < \dfrac{\varepsilon}{7}\cdot 7 = \varepsilon$. □

(e) Scratch: $\left|\dfrac{1}{x}-1\right| = \left|\dfrac{1-x}{x}\right| = \dfrac{|x-1|}{|x|}$. The dangerous factor is $\dfrac{1}{|x|}$, which is huge when x is near 0 — so we must keep x away from 0, and $|x-1|<1$ is *not* enough (it allows x close to 0). Take $|x-1|<\frac{1}{2}$ instead: then $\frac{1}{2}<x<\frac{3}{2}$, so $|x|>\frac{1}{2}$ and $\dfrac{1}{|x|}<2$. Then $\left|\dfrac{1}{x}-1\right|<2|x-1|$, which is $<\varepsilon$ when $|x-1|<\varepsilon/2$. Proof: Let $\varepsilon>0$; choose $\delta=\min(\frac{1}{2},\varepsilon/2)$. If $0<|x-1|<\delta$, then $x>\frac{1}{2}$, so $\dfrac{1}{|x|}<2$, and $\left|\dfrac{1}{x}-1\right| = \dfrac{|x-1|}{|x|} < 2|x-1| < 2\cdot\dfrac{\varepsilon}{2} = \varepsilon$. □ (Restricting $\delta\le\frac{1}{2}$ also keeps x in the domain of $1/x$.)

Problem 11 *Limits equal to $+\infty$, precisely.*

Recall: $\lim_{x\to a} f(x) = \infty$ means that for every $M>0$ there is $\delta>0$ such that $f(x)>M$ whenever $0<|x-a|<\delta$. For $x\to a^+$ replace "$0<|x-a|<\delta$" by "$a<x<a+\delta$" (and for $x\to a^-$ by "$a-\delta<x<a$"). Prove each statement from the definition: scratch work first, then the proof forward, starting with "Let $M>0$".

(a) $\lim_{x\to 3}\dfrac{1}{(x-3)^2} = \infty$. Which δ does your proof give for $M = 10{,}000$?

(b) $\lim_{x\to 2^+}\dfrac{3}{x-2} = \infty$.

(c) $\lim_{x\to 0^+}\dfrac{1}{\sqrt{x}} = \infty$. Which δ does your proof give for $M = 100$?

(d) $\lim_{x\to -1} \dfrac{5}{|x+1|} = \infty$.

(e) $\lim_{x\to 0}\left(1+\dfrac{1}{x^2}\right) = \infty$. (Hint: you may use $1+\dfrac{1}{x^2} > \dfrac{1}{x^2}$ — a lower bound that is easier to push above M.)

Solution. Now the challenge is a *height* M instead of a tolerance ε: whatever M is named, we must find a strip $0<|x-a|<\delta$ on which $f(x)>M$. The scratch work solves the inequality $f(x)>M$ for the distance to a; the proof runs forward and starts with "Let $M>0$". As M grows, δ shrinks: the higher the bar, the closer to a you must stand.

(a) Scratch: $\dfrac{1}{(x-3)^2} > M$. Both sides are positive, so taking reciprocals reverses the inequality: $(x-3)^2 < \dfrac{1}{M}$, i.e. $|x-3| < \dfrac{1}{\sqrt{M}}$. Proof: Let $M>0$. Choose $\delta = 1/\sqrt{M}$. If $0<|x-3|<\delta$, then $0<(x-3)^2<\delta^2 = \dfrac{1}{M}$, and taking reciprocals of positive numbers reverses the inequality: $\dfrac{1}{(x-3)^2} > M$. □ For $M = 10{,}000$: $\delta = 1/\sqrt{10{,}000} = 1/100 = 0.01$.

(b) Scratch: for $x>2$ the denominator is positive, and $\dfrac{3}{x-2} > M \iff x-2 < \dfrac{3}{M}$ (multiply by $x-2>0$, divide by $M>0$). Proof: Let $M>0$. Choose $\delta = 3/M$. If $2<x<2+\delta$, then $0<x-2<\dfrac{3}{M}$, so $\dfrac{3}{x-2} > \dfrac{3}{3/M} = M$. □

(c) Scratch: for $x>0$, $\dfrac{1}{\sqrt{x}} > M \iff \sqrt{x} < \dfrac{1}{M} \iff x < \dfrac{1}{M^2}$ (all quantities positive; squaring keeps the direction). Proof: Let $M>0$. Choose $\delta = 1/M^2$. If $0<x<\delta$, then $\sqrt{x} < \sqrt{\delta} = \dfrac{1}{M}$, so $\dfrac{1}{\sqrt{x}} > M$. □ For $M = 100$: $\delta = 1/100^2 = 10^{-4}$ — to push $1/\sqrt{x}$ above 100 you must come within one ten-thousandth of 0.

(d) Scratch: $\dfrac{5}{|x+1|} > M \iff |x+1| < \dfrac{5}{M}$ — the distance to -1 appears directly, on both sides of -1. Proof: Let $M>0$. Choose $\delta = 5/M$. If $0<|x+1|<\delta$, then $\dfrac{5}{|x+1|} > \dfrac{5}{\delta} = \dfrac{5}{5/M} = M$. □

(e) Scratch: solving $1+\dfrac{1}{x^2} > M$ exactly gives $\dfrac{1}{x^2} > M-1$, which only makes sense for $M>1$ — a case distinction we would rather avoid. Use a lower bound instead: $1+\dfrac{1}{x^2} > \dfrac{1}{x^2}$ always, so it is enough to make $\dfrac{1}{x^2} > M$, i.e. $|x| < \dfrac{1}{\sqrt{M}}$. Proof: Let $M>0$. Choose $\delta = 1/\sqrt{M}$. If $0<|x|<\delta$, then $x^2 < \dfrac{1}{M}$, so $\dfrac{1}{x^2} > M$, and therefore $1+\dfrac{1}{x^2} > \dfrac{1}{x^2} > M$. □ (General lesson: if $f(x)>M$ is awkward to solve, find a simpler $g \le f$ and push g above M.)

Problem 12 *Limits equal to $-\infty$, precisely.*

Write down what $\lim_{x\to a} f(x) = -\infty$ means in the same style (for every $N<0$ there is $\delta>0$ such that ...), and the one-sided versions. Then prove each statement from the definition. Start every proof with "Let $N<0$".

(a) $\lim_{x\to 0^+}\left(-\dfrac{1}{x}\right) = -\infty$.

(b) $\lim_{x\to 2} \dfrac{-3}{(x-2)^2} = -\infty$.

(c) $\lim_{x\to 1^-} \dfrac{2}{x-1} = -\infty$.

(d) $\lim_{x\to 0}\left(1-\dfrac{1}{|x|}\right) = -\infty$.

(e) $\lim_{x\to 3^+} \dfrac{x-5}{x-3} = -\infty$. (Hint: for $3 < x < 4$ the numerator is between -2 and -1; use that to find a simpler function *above* the given one.)

Solution. *The definition.* $\lim_{x\to a} f(x) = -\infty$ means: for every $N < 0$ there is $\delta > 0$ such that $f(x) < N$ whenever $0 < |x-a| < \delta$. (Whatever depth N is named, the graph is below it on a whole strip around a.) One-sided versions: for $x \to a^+$ replace "$0 < |x-a| < \delta$" by "$a < x < a+\delta$"; for $x \to a^-$ by "$a-\delta < x < a$". *Sign bookkeeping.* Since $N < 0$, write $|N| = -N > 0$; and remember that multiplying or dividing an inequality by a negative number *reverses* it.

(a) Scratch: for $x > 0$, $-\dfrac{1}{x} < N$. Multiply by -1 (reverses): $\dfrac{1}{x} > -N = |N|$. Both sides positive; reciprocals reverse again: $x < \dfrac{1}{|N|}$. Proof: Let $N < 0$. Choose $\delta = 1/|N|$. If $0 < x < \delta$, then $\dfrac{1}{x} > \dfrac{1}{\delta} = |N| = -N$, so $-\dfrac{1}{x} < N$. □

(b) Scratch: $\dfrac{-3}{(x-2)^2} < N \iff \dfrac{3}{(x-2)^2} > |N| \iff (x-2)^2 < \dfrac{3}{|N|} \iff |x-2| < \sqrt{3/|N|}$. Proof: Let $N < 0$. Choose $\delta = \sqrt{3/|N|}$. If $0 < |x-2| < \delta$, then $0 < (x-2)^2 < \delta^2 = \dfrac{3}{|N|}$, so $\dfrac{3}{(x-2)^2} > |N|$, and multiplying by -1: $\dfrac{-3}{(x-2)^2} < -|N| = N$. □

(c) Scratch: for $x < 1$ we have $x - 1 < 0$; write $\dfrac{2}{x-1} = -\dfrac{2}{1-x}$ with $1-x > 0$. Then $-\dfrac{2}{1-x} < N \iff \dfrac{2}{1-x} > |N| \iff 1-x < \dfrac{2}{|N|}$. Proof: Let $N < 0$. Choose $\delta = 2/|N|$. If $1-\delta < x < 1$, then $0 < 1-x < \dfrac{2}{|N|}$, so $\dfrac{2}{1-x} > |N|$, i.e. $\dfrac{2}{x-1} = -\dfrac{2}{1-x} < -|N| = N$. □ (A left-hand limit: the strip is $1-\delta < x < 1$, not $|x-1| < \delta$.)

(d) Scratch: $1 - \dfrac{1}{|x|} < N \iff -\dfrac{1}{|x|} < N-1 \iff \dfrac{1}{|x|} > 1-N = 1+|N| \iff |x| < \dfrac{1}{1+|N|}$. Proof: Let $N < 0$. Choose $\delta = \dfrac{1}{1+|N|}$. If $0 < |x| < \delta$, then $\dfrac{1}{|x|} > \dfrac{1}{\delta} = 1+|N|$, so $1 - \dfrac{1}{|x|} < 1-(1+|N|) = -|N| = N$. □

(e) Scratch: here the numerator is not constant, so we tame it first, as in the min trick. For $3 < x < 4$ we have $-2 < x-5 < -1$, in particular $x - 5 < -1$; dividing by $x - 3 > 0$ keeps the direction: $\dfrac{x-5}{x-3} < \dfrac{-1}{x-3}$. So it is enough to make $\dfrac{-1}{x-3} \le N$, i.e. $\dfrac{1}{x-3} \ge |N|$, i.e. $x - 3 \le \dfrac{1}{|N|}$. Two conditions on $x-3$: less than 1 and at most $1/|N|$; take $\delta = \min(1, 1/|N|)$. Proof: Let $N < 0$. Choose $\delta = \min(1, 1/|N|)$. If $3 < x < 3+\delta$, then $x < 4$, so $x - 5 < -1$; and $0 < x - 3 < \dfrac{1}{|N|}$, so $\dfrac{1}{x-3} > |N|$. Hence $\dfrac{x-5}{x-3} < \dfrac{-1}{x-3} < -|N| = N$. □

Problem 13 *A proof with a hole.*

A student "proves" that $\lim_{x\to 1}(3x+2) = 5$ as follows:

> *Let $\delta > 0$. Choose $\varepsilon = 3\delta$. If $0 < |x-1| < \delta$, then $|(3x+2)-5| = 3|x-1| < 3\delta = \varepsilon$. Therefore the limit is 5.*

Every inequality in it is correct. Why is it nevertheless not a proof? Rewrite it correctly.

Solution. *What is wrong.* The definition says: *for every* $\varepsilon > 0$ (chosen by the challenger) *there is* a $\delta > 0$ (chosen in response) such that $0 < |x-1| < \delta$ implies $|(3x+2)-5| < \varepsilon$. The order matters: ε comes first, and δ is allowed to depend on it. The student's text starts with "Let $\delta > 0$" and then *chooses* $\varepsilon = 3\delta$: the roles are reversed. What it proves is: for every δ there is *some* ε (namely 3δ) with $|f(x)-5| < \varepsilon$ on the strip — which is true, but says nothing about small ε; it only says that f is bounded near 1.

Why it cannot be right. With the same recipe one can "prove" that the limit is 6: let $\delta > 0$, choose

$\varepsilon = 3\delta + 1$; if $0 < |x-1| < \delta$ then $|(3x+2)-6| = |3x-4| = |3(x-1)-1| \le 3|x-1|+1 < 3\delta+1 = \varepsilon$. Every step is correct, and the conclusion is false — so the method is worthless.
The correct proof. Scratch: $|(3x+2)-5| = |3x-3| = 3|x-1|$, so we need $|x-1| < \varepsilon/3$. Proof: Let $\varepsilon > 0$. Choose $\delta = \varepsilon/3$. If $0 < |x-1| < \delta$, then $|(3x+2)-5| = 3|x-1| < 3\delta = \varepsilon$. Therefore $\lim_{x\to 1}(3x+2) = 5$. □

Challenge *(for those who finish early, or at home)*

(a) Prove from the definition that $\lim\limits_{x\to 0} x\sin\dfrac{1}{x} = 0$. (What δ works for a given ε?)

(b) Prove from the definition that $\lim\limits_{x\to 0}\dfrac{|x|}{x}$ does *not* exist. (Suppose it were some number L, take $\varepsilon = \frac{1}{2}$, and find two points within any δ of 0 whose values cannot both be within $\frac{1}{2}$ of L.)

Solution.

(a) Scratch: $\left|x\sin\dfrac{1}{x} - 0\right| = |x|\cdot\left|\sin\dfrac{1}{x}\right| \le |x|\cdot 1 = |x|$, because $|\sin(\text{anything})| \le 1$. So $|f(x)-0|$ is at most the distance $|x-0|$, and $\delta = \varepsilon$ should work. Proof: Let $\varepsilon > 0$. Choose $\delta = \varepsilon$. If $0 < |x| < \delta$, then $\left|x\sin\dfrac{1}{x}\right| \le |x| < \delta = \varepsilon$. Hence $\lim_{x\to 0} x\sin\dfrac{1}{x} = 0$. □ (This is the Squeeze Theorem for $-|x| \le x\sin\frac{1}{x} \le |x|$, written out with ε and δ.)

(b) Suppose, to the contrary, that $\lim_{x\to 0}\dfrac{|x|}{x} = L$ for some real number L. Apply the definition with the challenge $\varepsilon = \frac{1}{2}$: there is a $\delta > 0$ such that $\left|\dfrac{|x|}{x} - L\right| < \frac{1}{2}$ for every x with $0 < |x| < \delta$. Now pick two such points on opposite sides of 0: $x_1 = \delta/2$, where $\dfrac{|x_1|}{x_1} = 1$, and $x_2 = -\delta/2$, where $\dfrac{|x_2|}{x_2} = -1$. Both satisfy $0 < |x| < \delta$, so $|1-L| < \frac{1}{2}$ and $|-1-L| < \frac{1}{2}$. But then, by the triangle inequality,

$$2 = |1-(-1)| = |(1-L)+(L-(-1))| \le |1-L| + |L+1| < \tfrac{1}{2} + \tfrac{1}{2} = 1,$$

which is absurd ($2 < 1$). So no real number L can be the limit: $\lim_{x\to 0}\dfrac{|x|}{x}$ does not exist. □ (In words: within *any* δ of 0 the function takes the values 1 and -1, which are 2 apart; no single number is within $\frac{1}{2}$ of both.)